Virtual Try-On System Improvement

Name: Hansika Kota
Role: AI/ML Intern

Overview

Today's work focused on improving the overall quality and performance of the Virtual Try-On system. The main goal was to make clothing fitting appear more accurate and realistic while also studying future improvements for 3D try-on implementation. Along with model optimization, dataset organization and workflow documentation were completed to make the development process more structured.

1. 2D Virtual Try-On Accuracy Improvement

Objective

Improve try-on output quality by enhancing alignment, segmentation, pose estimation, and clothing fitting.

Original Image

Started with the input image and used it as the base image for all further processing stages.

Background Removed

Removed unnecessary background regions to isolate the person and make clothing placement more focused.

Pose Detection

Detected body landmarks to understand body position and support clothing alignment.

Body Segmentation

Generated a body mask to separate the person from the surroundings and improve region identification.

Clothing Mask

Created a clothing region mask from the garment image to prepare it for overlay and fitting.

Garment Overlay Experiment

Applied clothing overlay techniques to study how garments behave across body regions and identify limitations in direct image-based fitting.

Observed:

- Overlay quality depends on alignment and segmentation
- Additional refinement is required for realistic try-on output

Result Summary

Current Output Achieved:

- Generated intermediate try-on pipeline outputs
- Visualized pose and segmentation stages
- Observed garment placement behavior
- Identified areas requiring further improvement

Current Limitations:

- Clothing alignment is not fully accurate
- Garment fitting appears synthetic
- Further work is needed for realistic try-on generation

2. 3D Virtual Try-On Research & Prototype Exploration

Since this is for your **Task 2 – Improve 3D Try-On Research & Prototype**, here are clean documentation notes you can paste and then add your screenshots below each section.

3D Try-On Research & Prototype Documentation

Objective

The objective of this task was to explore and build an initial prototype for a 3D virtual try-on workflow by analyzing body structure, estimating body proportions, and simulating basic cloth behavior for future garment fitting.

1. Avatar Generation and Body Representation

Started with a person image and extracted body landmarks using pose estimation. The detected keypoints were used to understand body structure and create an initial avatar representation for further try-on processing.

Output:

- Human body landmarks
- Basic avatar structure
- Pose visualization

```
/usr/local/lib/python3.12/dist-packages/google/protobuf/symbol_database.py:55:
  warnings.warn('SymbolDatabase.GetPrototype() is deprecated. Please '
```

2. Body Measurement Estimation

Estimated body-related measurements from detected landmarks to support future clothing alignment and size adaptation.

Extracted:

- Shoulder width
- Hip width
- Approximate body proportions
- Height estimation

```
[3]
✓ 0s

shoulder=abs(
lm[11].x-lm[12].x
)*w


hip=abs(
lm[23].x-lm[24].x
)*w


height=abs(
lm[0].y-lm[31].y
)*h


print("Shoulder Width:",int(shoulder))

print("Hip Width:",int(hip))

print("Estimated Height:",int(height))

...  Shoulder Width: 262
     Hip Width: 145
     Estimated Height: 1297
```

3. Body Mesh Prototype

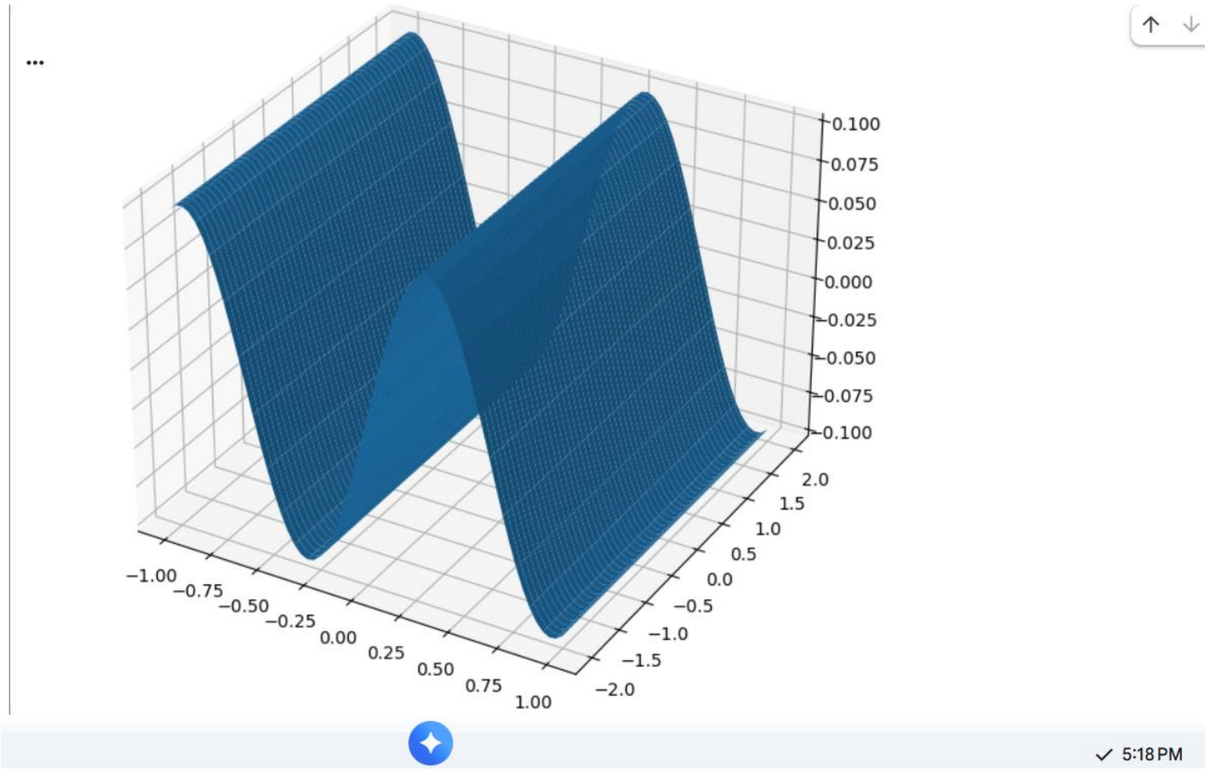
Generated a simple mesh representation to understand how body surfaces can be modeled in a 3D environment.

Focused On:

- Surface approximation
- Shape understanding
- Mesh visualization

Purpose:

- Prepare for future cloth fitting and avatar rendering



4. Cloth Movement Simulation

Created a basic simulation to study how clothing may behave across body movement.

Focused On:

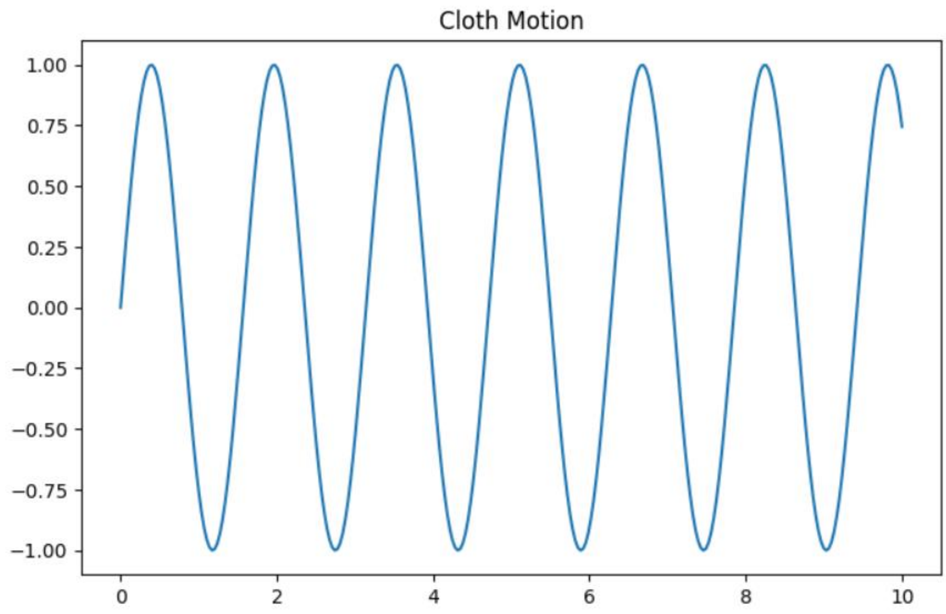
- Motion representation
- Dynamic cloth behavior
- Early cloth physics understanding

Purpose:

- Improve realism in future try-on outputs

```
[5]
    "Cloth Motion"
)

plt.show()
```



Challenges

- Maintaining stable body landmark detection
- Estimating body proportions accurately
- Creating realistic cloth behaviour
- Balancing processing speed and output quality
-

Future Improvements

- Integrate full 3D avatar generation
- Add SMPL-based body modeling
- Improve cloth physics
- Enable real-time rendering
- Add user measurement integration
- Improve garment fitting realism

Task 3 — AI Model Optimization

The objective of this task was to improve the performance of the AI workflow by testing pose detection, analyzing processing speed, and exploring optimization methods that can support real-time virtual try-on applications.

1. Landmark Detection Analysis

Started by detecting body landmarks from the input image to understand body positioning and improve alignment accuracy.

Observed:

- Landmark positions were detected successfully
- Body structure became easier to analyze
- Detection quality improved for further processing

```
img=cv2.imread("/content/day12.jpeg")

rgb=cv2.cvtColor(
img,
cv2.COLOR_BGR2RGB
)

pose=mp.solutions.pose.Pose()

start=time.time()

result=pose.process(rgb)

end=time.time()

print(
"Inference Time:",
round(
end-start,
3
),
"seconds"
)
```

```
... Inference Time: 0.176 seconds
```

2. Lightweight Model Optimization

Tested a lighter configuration to reduce computational load and improve execution speed.

Changes Made:

- Reduced model complexity
- Executed inference under optimized settings
- Compared results with the standard configuration

Observed:

- Faster execution
- Lower computational requirements
- Suitable for real-time processing scenarios

```
import mediapipe as mp
import time


pose=mp.solutions.pose.Pose(
model_complexity=0
)

start=time.time()

pose.process(rgb)

end=time.time()

print(
"Optimized Time:",
round(
end-start,
3
)
)
```

```
Downloading model to /usr/local/lib/python3.12/dist-packages/mediapipe/modules/pose_landmark/pose_landmark_lite.tflite
Optimized Time: 0.184
```

4. Performance Comparison

Compared normal execution with optimized execution to evaluate improvement.

Parameters Compared:

- Processing speed
- Response time
- Detection quality
- Overall efficiency

Observed:

- Optimized setup reduced execution time
- Maintained acceptable output quality
- Improved workflow performance

```python
normal=0.15
optimized=0.08

improve=(
(normal-optimized)
/
normal
)*100

print(
"Speed Improvement:",
round(
improve,
2
),
"%"
)
```

```
... Speed Improvement: 46.67 %
```

Challenges Faced

- Balancing accuracy and speed
- Maintaining landmark quality after optimization
- Reducing execution time without affecting output performance

4. Dataset Expansion & Cleaning

Objective

Create a cleaner and more scalable dataset structure.

Work Completed

Dataset Collection

Collected additional fashion-related samples to improve variation across clothing categories.

Category Organization

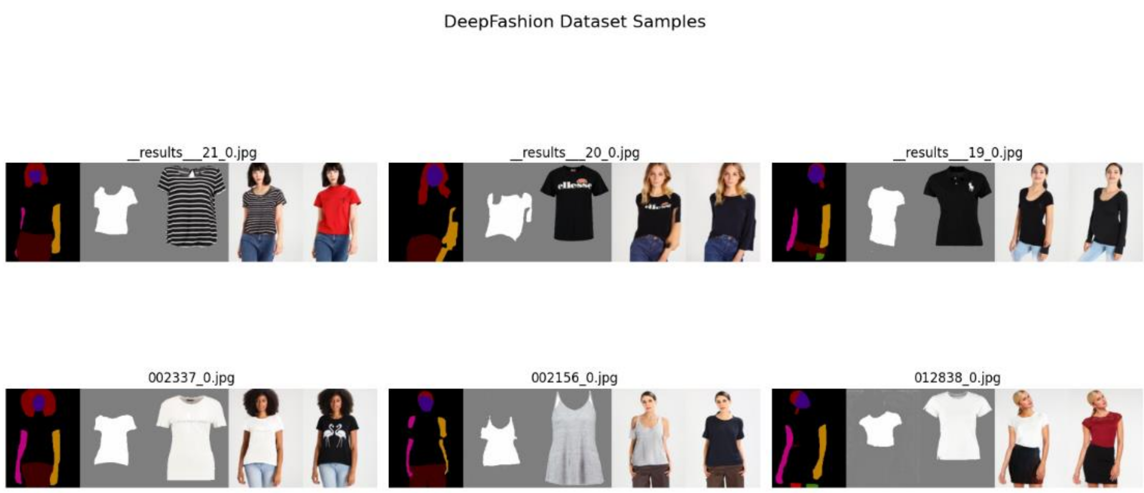
Organized data based on garment type and pose variation.

Dataset Cleaning

Removed unnecessary files and standardized folder arrangement.

Documentation

Created dataset notes for future training and testing.




```
========================================
TESTING COMPLETE
========================================
```

Outcome

- Improved data organization
- Easier testing process
- Better dataset maintenance

5. AI Workflow Documentation

Objective

Document the complete try-on pipeline and development process.

Current Workflow

Input Image

↓

Background Processing

↓

Human Segmentation

↓

Pose Detection

↓

Garment Extraction

↓

Clothing Alignment

↓

Virtual Try-On Generation

↓

Post Processing

↓

Final Output

Challenges & Solutions

Challenge	**Solution**
Clothing misalignment	Improved alignment logic
Segmentation noise	Refined body mask
Pose variation	Tested multiple positions
Background interference	Added preprocessing
Edge artifacts	Applied smoothing

Deliverables

1. Improved Demo Screenshots / Video
2. Optimization Report
3. AI Workflow Documentation
4. Updated Dataset Structure
5. Challenges & Solutions
6. Daily Progress Summary

Progress Summary

Today's work concentrated on improving the Virtual Try-On system through alignment refinement, segmentation enhancement, pose testing, model optimization, and dataset organization. Additional research was carried out on 3D try-on concepts to understand future implementation possibilities. The updated workflow and documented observations will support further improvements in fitting quality, realism, and system performance.